

ECG Arrhythmia Classification

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Abstract—Electrocardiogram (ECG) interpretation is a critical tool for detecting cardiac abnormalities and assisting clinical decision-making. This document addresses the arrhythmia classification problem by developing and optimizing a machine learning model. The ECG dataset that was used contains 32 different numerical interpretations of ECG, unique number of each patient, and 5 different heartbeat classes. After establishing the initial classification model using logistic regression, ...

Keywords—machine learning, logistic regression, softmax, cross entropy, electrocardiogram (ECG), Exploratory Data Analysis (EDA), Principal Component Analysis (PCA)

I. INTRODUCTION

Cardiovascular diseases remain one of the important health concerns, and abnormalities in cardiac rhythm, known as arrhythmias, can be found through the analysis of electrocardiogram (ECG) signals. ECG recordings contain information about the electrical activity of the heart and can show changes in cardiac rhythm and morphology. However, manually analyzing large quantities of ECG recordings can be time-consuming and the development of computational methods to identify different heartbeat patterns can be useful.

Machine learning provides a time-efficient analysis by using ECG-derived features to automatically classify heartbeats according to their characteristics. In this project, a multiclass classification model is developed to distinguish between five different heartbeat categories using features extracted from ECG recordings. To establish a working classification model, various methods were used, including data preparation, exploratory data analysis, dimensionality reduction, model implementation, and evaluation.

II. OBJECTIVE

The objective of this project is to develop a multiclass machine-learning model to classify ECG heartbeats into five categories: Normal (N), Supraventricular Ectopic Beat (SVEB), Ventricular Ectopic Beat (VEB), Fusion Beat (F), and Unknown Beat (Q). The project additionally aims to analyze the characteristics and limitations of the dataset, implement the classification algorithm without relying on machine learning frameworks, evaluate its ability to generalize to untrained ECG recordings, and improve the model using a machine learning framework.

III. BACKGROUND AND THEORETICAL FOUNDATIONS

To understand the basic background and theoretical foundations of the arrhythmia classification problem using ECG analysis, a few concepts that are better to know are addressed in the following sections.

A. Electrocardiogram (ECG)

An electrocardiogram (ECG) is a recording of the electrical activity produced by the heart during each cardiac cycle. A typical heartbeat produces a recognizable waveform consisting primarily of the P wave, QRS complex, and T wave. These components represent different stages of electrical activation and recovery of the heart.

The electrical cycle begins with the P wave, which represents atrial depolarization. During this stage, an electrical signal spreads through the atria, causing them to contract and move blood into the ventricles.

After the P wave, the signal travels toward the ventricles and produces the QRS complex, which represents ventricular depolarization. This electrical activation causes the ventricles to contract and pump blood out of the heart. The QRS complex consists of three possible deflections:

- Q wave: The first negative deflection before the main positive peak. It represents the early stage of ventricular depolarization.
- R wave: The main positive deflection and usually the most prominent peak in an ECG heartbeat. It represents a major portion of ventricular depolarization. Because R peaks are generally easy to identify, the distance between consecutive R peaks, known as the RR interval, is commonly used to analyze heart rate and rhythm.
- S wave: The negative deflection following the R wave. It represents the later stages of ventricular depolarization.

Together, the Q, R, and S waves form the QRS complex. The duration and shape, or morphology, of this complex provide important information about how electrical signals travel through the ventricles. Abnormal heartbeats can produce QRS complexes with different widths, amplitudes, or shapes, making QRS-related features particularly useful for arrhythmia classification.

Following the QRS complex is the T wave, which represents ventricular repolarization. During this process, the ventricular cells electrically recover and return toward their resting state in preparation for the next heartbeat.

A simplified representation of one ECG cycle can therefore be described as:

P wave → QRS complex → T wave

or physiologically as:

Atrial activation → Ventricular activation → Ventricular recovery

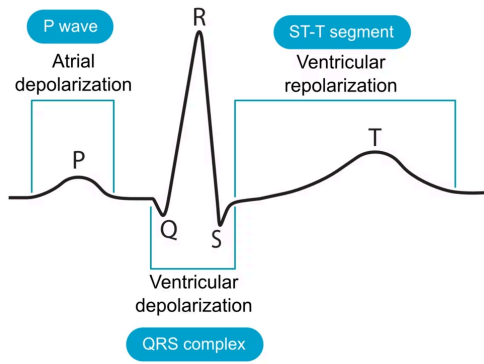


Image 1. Electrocardiogram basic components ^[1]

In addition to the individual waves, ECG analysis considers the intervals between these components. These intervals describe the timing of electrical conduction through different regions of the heart. For example, the RR interval describes the timing between consecutive heartbeats, while the PQ, QRS, QT, and ST measurements provide information about different stages of electrical conduction, ventricular activation, and recovery.

Because arrhythmias can alter the timing, amplitude, duration, and morphology of these ECG components, measurements derived from the P, Q, R, S, and T waves can be used as input features for machine-learning models designed to distinguish normal and abnormal heartbeats.

IV. DATASET AND DATA PREPARATION

A. Dataset Source and Feature Extraction

Before getting into analyzing the dataset, it is important to mention that the ECG features used in this study^[2] were not extracted directly as part of the proposed machine-learning pipeline. Instead, the dataset provides features previously extracted from the original MIT-BIH Arrhythmia Database^[3] according to the feature-extraction methodology presented by Sakib, Fouda, and Fadlullah^[4]. Their methodology consists of ECG preprocessing, heartbeat segmentation, peak detection, and feature extraction. The resulting features are organized into four principal categories: RR interval features, heartbeat interval features, heartbeat wave-amplitude features, and QRS morphology features. Measurements are extracted from the two ECG channels available in the original recordings, providing two sets of features for each heartbeat. These numerical features are subsequently used as the input variables for the classification models developed in this study.

B. Dataset analysis

The dataset contains 100,689 heartbeat instances and 34 columns, including the ECG-derived features, the heartbeat classification label (type), and the source ECG recording identifier (record). Table 1 summarizes the physiological or computational meaning of the extracted features and Table 2 explains the heartbeat classes (type) and the source identifier(record).

Feature	Meaning	Description / Relevance
pre-RR	Previous RR interval	Time interval between the R peak of the previous heartbeat and the R peak of the current heartbeat. It describes the timing of the current beat relative to the preceding beat and can help identify premature or irregular beats.
post-RR	Following RR interval	Time interval between the R peak of the current heartbeat and the R peak of the following heartbeat. Together with pre-RR, it provides information about local heart rhythm and irregularities.
pPeak	P-wave peak amplitude	Amplitude of the P wave, which represents atrial depolarization. Variations in its amplitude and morphology may indicate changes in atrial electrical activity.
qPeak	Q-wave peak amplitude	Amplitude of the Q wave, the initial negative deflection of the QRS complex. It contributes information about the morphology of ventricular depolarization.
rPeak	R-wave peak amplitude	Amplitude of the R wave, typically the largest positive deflection of the QRS complex. It provides information about ventricular electrical activity and QRS morphology.
sPeak	S-wave peak amplitude	Amplitude of the S wave, the negative deflection following the R wave. It provides additional information about the shape of the QRS complex.
tPeak	T-wave peak amplitude	Amplitude of the T wave, which represents ventricular repolarization. Changes in the T wave can reflect variations in ventricular recovery.
qrs_inter val	QRS duration	Duration of the QRS complex. It represents the time associated with ventricular depolarization and is useful for distinguishing beats with abnormal ventricular conduction or morphology.

pq_inter val	PQ interval	Time interval associated with conduction from atrial depolarization toward the beginning of ventricular depolarization. It provides information about atrioventricular conduction.
qt_interv al	QT interval	Time from the beginning of ventricular depolarization to the completion of ventricular repolarization. It represents the overall duration of ventricular electrical activity.
st_interv al	ST interval	Interval associated with the period between ventricular depolarization and repolarization. It provides information about the transition from the QRS complex to the T wave.
qrs_mor ph0	QRS morpholo gy value 0	A sampled/derived value representing part of the morphology (shape) of the QRS complex.
qrs_mor ph1	QRS morpholo gy value 1	A second value used to characterize the shape of the QRS complex.
qrs_mor ph2	QRS morpholo gy value 2	A third value describing QRS morphology.
qrs_mor ph3	QRS morpholo gy value 3	A fourth value describing QRS morphology.
qrs_mor ph4	QRS morpholo gy value 4	A fifth value describing QRS morphology. Together, the morphology features allow the model to distinguish different QRS shapes.

Table 1. Description of each ECG feature

Column	Meaning	Role
record	MIT-BIH ECG record identifier	Identifies the source ECG recording from which the heartbeat was extracted. It should generally be treated as an identifier/grouping variable

		rather than a predictive physiological feature.
type	Heartbeat class	Target variable indicating the heartbeat/arrhythmia category to be predicted by the classification model.

Table 2. Description of type and record

C. ECG Channel Representation

The dataset contains ECG measurements obtained from two different channels, represented by the prefixes 0_ and 1_. Both channels record the electrical activity of the same heartbeat, but from different ECG signals, providing different representations of the heart's electrical activity. Consequently, the same set of features is extracted from each channel. For example, 0_rPeak represents the R-wave peak measured from Channel 0, while 1_rPeak represents the corresponding measurement from Channel 1. Using two channels can provide complementary information because the amplitude and morphology of an ECG waveform can appear differently depending on the signal from which it is measured. Therefore, the dataset contains two sets of the same ECG features, allowing the classification model to use information from both channels when distinguishing between different heartbeat types. The following Table explains the data type, observed range and the measurement unit for each ECG feature.

Feature	Type	Channel 0 Range	Channel 1 Range	Unit
pre-RR	Integer	11 – 2,114	11 – 2,114	Extracted interval
post-RR	Float	6 – 2,114	6 – 2,114	Extracted interval
pPeak	Float	−3.094 – 3.821	−2.108 – 5.161	Signal amplitude
qPeak	Float	−3.871 – 3.081	−5.482 – 4.034	Signal amplitude
rPeak	Float	−2.373 – 4.019	−3.847 – 4.034	Signal amplitude
sPeak	Float	−3.623 – 4.019	−6.155 – 3.318	Signal amplitude
tPeak	Float	−2.006 – 4.118	−2.493 – 7.003	Signal amplitude

qrs_inter val	Integer	0 – 170	0 – 260	Extracted interval
pq_inter val	Integer	1 – 150	1 – 185	Extracted interval
qt_inter val	Integer	5 – 334	5 – 412	Extracted interval
st_interv al	Integer	1 – 141	1 – 215	Extracted interval
qrs_mor ph0	Float	−3.871 – 3.081	−5.482 – 4.034	Signal amplitude
qrs_mor ph1	Float	−2.629 – 2.870	−4.494 – 3.586	Signal amplitude
qrs_mor ph2	Float	−2.840 – 3.331	−4.398 – 2.692	Signal amplitude
qrs_mor ph3	Float	−2.995 – 3.225	−4.070 – 3.756	Signal amplitude
qrs_mor ph4	Float	−3.086 – 2.633	−4.520 – 2.869	Signal amplitude

Table 3. Data types, observed ranges, and measurement units of the ECG features used in the dataset.

The features in the dataset are represented primarily through extracted intervals and signal-amplitude values. An extracted interval describes the distance between two characteristic points of the ECG waveform. Since an ECG is a digital signal, it consists of a sequence of individual measurements called samples. The original MIT-BIH Arrhythmia Database was recorded at a sampling frequency of 360 Hz, meaning that 360 signal samples were collected per second. Therefore, distances between detected ECG points can be represented as the number of samples separating them. Features such as the RR, PQ, QT, ST, and QRS intervals belong to this category and provide information about the timing of cardiac electrical activity.

Signal amplitude, on the other hand, represents the magnitude of the ECG signal at a particular point. In an ECG waveform, the horizontal axis represents time or sample position, while the vertical axis represents signal amplitude. Positive and negative values indicate the direction and magnitude of the signal relative to its baseline. The P, Q, R, S, and T peak features use these amplitude values to characterize important parts of the heartbeat waveform. The QRS morphology features similarly contain values derived from the ECG signal to describe the shape of the QRS complex. Since these values were obtained after ECG preprocessing and feature extraction, they are treated as processed signal-amplitude values rather than assumed to represent direct clinical measurements in millivolts.

D. Target Variable and Record Identifier

In addition to the extracted ECG features, the dataset contains two columns with specific roles in the classification pipeline: type and record. The type column represents the target variable and assigns each heartbeat to one of five categories: Normal (N), Supraventricular Ectopic Beat (SVEB), Ventricular Ectopic Beat (VEB), Fusion Beat (F), or Unknown Beat (Q). Since each heartbeat is assigned to a single category, the task is formulated as a mutually exclusive multiclass classification problem.

The record column identifies the source ECG recording from which each heartbeat was extracted. It is not considered a predictive ECG feature and is therefore not provided as an input to the classification model. However, it is retained during data preparation to construct the training and test sets at the recording level, as described in the preprocessing section.

V. EXPLORATORY DATA ANALYSIS

A. Data Quality Assessment

Prior to modeling, the dataset was examined for missing values, duplicated observations, and abnormalities in the numerical features. As a result, it is discovered that there are not any missing or duplicated values. However, there were two different sets of features that contained the exact same values, which were 0_pre-RR with 1_pre-RR and 0_post-RR with 1_post-RR. Based on the observations, two new features called pre-RR and post-RR were created, with the following deletion of the features: 0_pre-RR, 1_pre-RR, 0_post-RR, and 1_post-RR.

B. Target Class Distribution

Based on the distribution of the 5 distinct classes, it was found that there is an imbalance of the distribution between the classes. The Normal(N) heartbeat accounts for 89.4% of the classes, meanwhile Unknown beat(Q) comprises approximately 0.014 %. Ventricular Ectopic Beat (VEB) constitutes 6.9%, Supraventricular Ectopic Beat (SVEB) forms 2.7% , and Fusion Beat (F) makes up 0.79% of the target data. This imbalance can affect the training process because the model is exposed to significantly more examples from the majority class. This can result that the optimization process may favor correctly classifying Normal (N) heartbeats while giving less importance to errors involving minority classes. As a result, the model may achieve high overall accuracy while still performing poorly at detecting less frequent arrhythmia classes. Therefore, evaluation metrics such as precision, recall, F1-score, and macro-F1 are considered in addition to accuracy when evaluating the classification model.

C. Feature Distribution Analysis

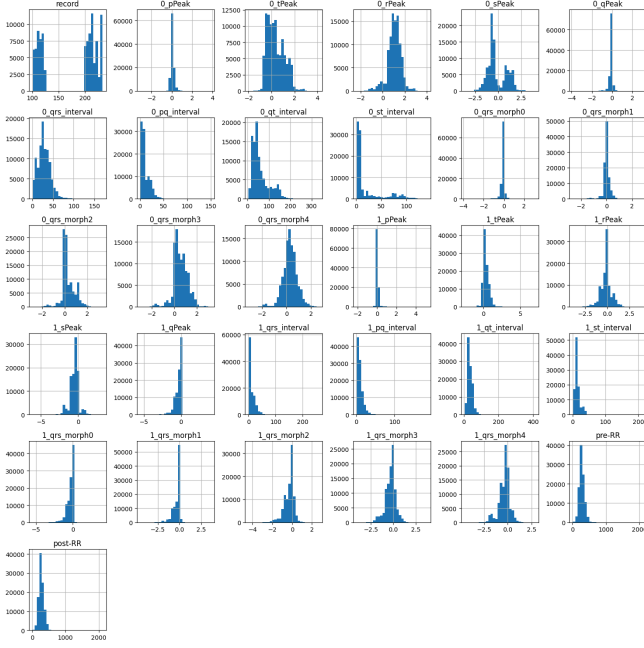


Image 2. Feature distributions

The feature distributions based on the Image 2 show differences in scale, skewness, variability, and distribution shape. Interval-based features generally exhibit positively skewed distributions and wider numerical ranges, whereas amplitude and morphology features are concentrated within smaller ranges. Some features also exhibit multiple regions of concentration, which may reflect differences among heartbeat classes. These observations motivated feature standardization before training scale-sensitive models, while extreme observations were retained because they may represent genuine cardiac variability.

D. Feature Correlation and Redundancy

A correlation analysis was performed to identify relationships and potential redundancy among the numerical features. The correlation matrix showed several strong relationships, particularly among the QRS morphology features. Notably, 0_qPeak and 0_qrs_morph0 , as well as 1_qPeak and 1_qrs_morph0 , showed perfect positive correlations. Therefore, the two qrs_morph0 features were removed while the corresponding $qPeak$ features were retained due to their clearer physiological interpretation.

VI. DATA PREPROCESSING

A. Record-Level Train-Test Split

The record variable identifies the ECG recording from which each heartbeat was extracted. Since multiple heartbeat samples can originate from the same recording, randomly splitting individual samples could place heartbeats from the same recording in both the training and test sets. These samples may share recording-specific characteristics, potentially resulting in an overly optimistic evaluation of model performance.

Therefore, the dataset was split at the record level, assigning approximately 80% of the records to the training set and 20% to the test set. This ensures that records used for testing were not observed during training and provides a more rigorous evaluation of the model's ability to generalize to unseen ECG recordings. The record variable was used only for the data split and was excluded from the model's input features.

B. Feature Standardization

Standardization was performed because the ECG features were represented on substantially different numerical scales like we observed during the feature distribution analysis. Each feature was transformed to have approximately zero mean and unit standard deviation. To prevent information leakage, the mean and standard deviation were calculated from the training data. The same training-derived parameters were subsequently used to transform the test data to standardize the transformation of the data.

C. Principal Component Analysis

Principal Component Analysis (PCA) was applied after standardization to investigate whether the dimensionality of the feature space could be reduced while preserving most of its variability. PCA was calculated exclusively from the standardized training data, and the resulting principal component transformation was subsequently applied to the test data.

A cumulative explained variance threshold of 95% was selected. The first 14 principal components explained approximately 94.76% of the total variance, while 15 components explained approximately 96.11%. Therefore, 15 principal components were retained, representing the smallest number of components exceeding the selected 95% threshold.

VII. CLASSIFICATION METHODOLOGY

A. Logistic regression

Logistic regression was selected as the initial model. Logistic regression provides a simple and interpretable approach to classification and gives insights of whether more complex models or preprocessing strategies are needed subsequently. Beginning with a linear classifier also makes it possible to determine how effectively the extracted ECG features separate the heartbeat classes without introducing additional model complexity.

B. Multiclass Output and Softmax

For multinomial logistic regression, the linear output of the model is calculated as:

$$Z = WX + b$$

X is the input matrix that contains all the heartbeat samples used by the model. Each row represents one heartbeat, while each column represents one input feature. After PCA, each heartbeat is represented by 15 principal components.

W is the weight matrix that contains the weights learned by the logistic regression model during training. Since there are

five possible heartbeat classes, the model learns a separate set of 15 weights for each class.

b is the bias vector that contains one additional bias value for each of the five classes. These values are also learned during training.

Lastly, Z is the logit matrix that contains the scores calculated by the model. For every heartbeat, the model produces five scores, one for each possible class. These scores are then converted into probabilities using the softmax function.

$$\hat{y}_{ik} = \frac{e^{z_{ik}}}{\sum_{j=1}^K e^{z_{ij}}}$$

Here, Softmax was selected as the output activation function because the ECG classification problem consists of five mutually exclusive classes: N, SVEB, VEB, F, and Q. Each heartbeat is assigned to only one of these classes. Softmax converts the model's output scores into a probability distribution across all five classes, where the probabilities sum to one. This allows the class with the highest predicted probability to be selected as the final prediction. In contrast, applying sigmoid independently to each output would treat each class as an independent binary classification problem, allowing multiple classes to receive high probabilities simultaneously, which is not appropriate for this mutually exclusive classification task.

C. Cross-Entropy Loss

$$L = -\frac{1}{m} \sum_{i=1}^m \sum_{k=1}^K y_{ik} * \log(\hat{y}_{ik})$$

Categorical cross-entropy was used as the loss function to quantify the difference between the predicted class probabilities and the true one-hot encoded labels. The loss penalizes the model when a low probability is assigned to the correct heartbeat class and provides the optimization objective for training.

D. Gradient Descent

To train the logistic regression model, gradient descent was used to minimize the categorical cross-entropy loss. The objective of this optimization process is to determine the weights and biases that minimize the difference between the predicted class probabilities and the actual heartbeat classes.

$$\frac{\partial L}{\partial z} = \hat{y} - y$$

The gradients represent how much the loss changes with respect to each model parameter. The difference between the predicted probabilities and the true labels is first calculated and then used to obtain the gradients for the weight matrix and bias vector. These gradients indicate both the direction and magnitude in which the parameters should be adjusted to reduce the loss.

$$\frac{\partial L}{\partial W} = \frac{1}{m} X^T (\hat{y} - y)$$

$$\frac{\partial L}{\partial b} = \frac{1}{m} \sum_{i=1}^m (\hat{y}_i - y_i)$$

The weights and biases are updated in the opposite direction of their respective gradients because the objective is to minimize the loss function. The learning rate, in this case set as 0.01, controls the size of each update. This process is repeated over multiple epochs, allowing the model parameters to progressively adjust toward values that produce a lower classification loss.

$$W \leftarrow W - \alpha \frac{\partial L}{\partial W}$$

$$b \leftarrow b - \alpha \frac{\partial L}{\partial b}$$

Batch gradient descent was selected for the baseline model because it provides a straightforward implementation of the optimization process. In each epoch, all training observations are used to calculate the gradients before performing a single parameter update. This produces a stable estimate of the gradient without the fluctuations. Furthermore, the relatively small number of model parameters after PCA made full-batch optimization computationally feasible for this experiment.

Therefore, during each epoch, the complete training dataset was passed through the model, the cross-entropy loss and corresponding gradients were calculated, and the weights and biases were updated once. This procedure was repeated for 1,000 epochs.

VIII. EVALUATION METRICS

To evaluate the performance of the classification model, several metrics were considered. Since the dataset presents a class imbalance, overall accuracy alone may not adequately represent the model's ability to classify the less frequent heartbeat categories. Therefore, accuracy was complemented with precision, recall, F1-score, macro-F1, and a confusion matrix.

A. Accuracy

$$Accuracy = \frac{Correct\ predictions}{Total\ predictions}$$

Accuracy represents the proportion of heartbeat samples that are correctly classified out of the total number of samples. Although it provides a general measure of model performance, it can be misleading for imbalanced datasets. In this dataset, Normal (N) heartbeats represent the majority of observations; therefore, a model could obtain high accuracy by correctly predicting the N class while performing poorly on the minority arrhythmia classes.

B. Precision

$$Precision = \frac{TP(True\ positive)}{TP + FP(False\ Positive)}$$

Precision measures how many of the samples predicted as a particular class actually belong to that class. A high precision indicates that when the model predicts a specific heartbeat category, the prediction is usually correct.

C. Recall

$$Recall = \frac{TP}{TP + FN(False\ Negative)}$$

Recall measures the proportion of samples belonging to a particular class that the model successfully identifies. This metric is particularly important in an imbalanced classification problem because it reveals whether the model is failing to detect observations from minority classes.

D. F1-Score

$$F1 = 2 \frac{Precision * Recall}{Precision + Recall}$$

The F1-score combines precision and recall into a single metric using their harmonic mean. A high F1-score therefore requires both high precision and high recall. This makes the F1-score useful for evaluating each heartbeat class when the class distribution is imbalanced.

E. Macro-F1

$$Macro\ F1 = \frac{1}{K} \sum_{k=1}^K F1_k$$

Macro-F1 is calculated by obtaining the F1-score of each class independently and then calculating their average. Unlike accuracy, each heartbeat category contributes equally to macro-F1 regardless of the number of observations belonging to that class. For this reason, macro-F1 is particularly useful in this study for determining whether the model performs consistently across both majority and minority classes.

F. Confusion Matrix

A confusion matrix was also used to examine the specific classification errors made by the model. Each row represents the actual heartbeat class, while each column represents the class predicted by the model. Values along the main diagonal correspond to correct classifications, whereas values outside the diagonal represent misclassifications. This allows the model's behavior across individual heartbeat categories to be analyzed in greater detail than with a single performance metric.

IX. BASELINE MODEL RESULTS

A. Training and Test Accuracy

The baseline multinomial logistic regression model achieved a training accuracy of 91.79% and a test accuracy

of 95.51%. At first, these results appear to indicate strong classification performance. However, considering the substantial class imbalance identified during EDA, accuracy alone is insufficient to determine whether the model successfully distinguishes all five heartbeat categories. Therefore, the confusion matrix and class-specific performance metrics were analyzed.

B. Confusion Matrix

The confusion matrix reveals that the high test accuracy is primarily associated with the model's performance on the majority N class. From the 21,654 N observations in the test set, 21,644 were correctly classified. In contrast, all 373 SVEB observations and all 26 F observations were incorrectly classified as N. For VEB, the model correctly classified 204 observations but incorrectly classified 618 as N. This indicates that the baseline model has a strong tendency to predict the majority class.

Additionally, the selected record-level test split contained no Q observations. Consequently, the performance of the model on the Q class cannot be evaluated using this particular split. This limitation indicates that the record-level splitting procedure should be revised to ensure representation of all classes in the evaluation set.

C. Per-Class Performance

Class	Precision	Recall	F1-score
N	0.9551	0.9995	0.9768
SVEB	0.0000	0.0000	0.0000
VEB	0.9533	0.2482	0.3938
F	0.0000	0.0000	0.0000
Q	—	—	—

Table 3. Class Performance Table

The class-specific metrics further demonstrate the limitations of the baseline model. The N class achieved an F1-score of 0.9768, indicating great performance for the majority class. VEB achieved a high precision of 0.9533, meaning that VEB predictions were generally correct; however, its recall was only 0.2482, indicating that the model detected only approximately one quarter of the actual VEB observations. SVEB and F both obtained precision, recall, and F1-scores of zero because the model did not correctly classify any observations from these categories.

D. Macro-F1 and Overall Baseline Assessment

Despite achieving a test accuracy of 95.51%, the model obtained a macro-F1 score of only 0.2741. The difference

between these two metrics proves the effect of class imbalance on the baseline classifier. Since most observations belong to the N class, strong performance on this class produces high overall accuracy even when the model performs poorly on several minority classes.

Therefore, the baseline model cannot yet be considered effective for balanced multiclass heartbeat classification. These results require further experimentation with methods designed to reduce the influence of class imbalance and improve the detection of minority heartbeat categories.

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